

FlowMES

Manufacturing Execution System — Complete Walkthrough

Architecture, every module, where each module's data comes from in a real factory, and how to connect it.

Live: flow-mes.vercel.app | API: flowmes-production.up.railway.app

1. What FlowMES is, and where it starts

FlowMES is a Manufacturing Execution System (MES) for small and mid-sized manufacturers. It puts the whole factory — machines, materials, production, quality, maintenance and live machine data — into one web dashboard the team can use from any device, replacing spreadsheets, WhatsApp updates and guesswork.

The stack

Frontend — a Next.js web app (hosted on Vercel) that runs in the browser. **Backend** — a FastAPI service (hosted on Railway) that holds all the logic and talks to the database. **Database** — PostgreSQL, the single source of truth. **Live data** — MQTT for machine telemetry and a WebSocket for pushing live updates to the dashboard.

Where a session starts (the request flow)

1. You open **flow-mes.vercel.app** and sign in. 2. The backend checks your password (bcrypt) and returns a signed token (JWT) that carries your **role** (Admin / Supervisor / Operator) and your **company**. 3. The dashboard loads and asks the backend for data — machines, work orders, inventory, analytics, and so on. 4. The backend reads PostgreSQL and returns it. 5. A live WebSocket keeps the dashboard moving in real time. Every screen you see is the dashboard asking the backend a question and drawing the answer.

Two kinds of data — this is the key idea

(A) Human / system data — work orders, inventory, quality checks, purchase orders, maintenance. In a real factory this is entered by people (operators, supervisors, store, QC) or imported from an existing system (Tally, an ERP, Excel). FlowMES then keeps it live and connected.

(B) Machine / sensor data — temperature, vibration, RPM, power, run status, part counts. This comes straight off the machines and their PLCs through a small **edge agent** on your shop floor (explained in Section 4). No one types this in.

2. Core MES (included in every plan)

Real-time machine monitoring — the foundation every other module builds on.

Overview

What it does. The top-level pulse of the plant: running machines, breakdowns, average utilization and OEE, total downtime, shift efficiency and the top downtime reason.

Where the data comes from (real factory). It doesn't have its own input — it aggregates everything from the modules below.

How to connect. Automatic. As soon as machines, downtime, shifts and production records exist, this fills in.

Machines

What it does. A register of every machine with its live status (Running / Idle / Breakdown / Maintenance) and utilization %.

Where the data comes from (real factory). You set up the machine list once. The live status comes from each machine's PLC/sensors via the edge agent, or from a manual status update by a supervisor.

How to connect. Add machines on this tab. Live status then flows in from the Connectivity / IoT layer, or the edge agent updates a machine's status automatically.

Downtime

What it does. A log of every stoppage with its reason, duration and notes — and the management report's 'top loss reason'.

Where the data comes from (real factory). Operators record why a machine stopped; or the system auto-creates an entry the moment a machine flips to Breakdown.

How to connect. Manual entry on the Downtime tab, or automatic from machine status changes / PLC fault signals.

Shifts

What it does. Per-shift target vs actual output and the resulting efficiency %.

Where the data comes from (real factory). The shift supervisor enters actual output, or it rolls up automatically from operator/production counts.

How to connect. Manual entry, or derived from production records.

Analytics

What it does. OEE broken into Availability x Performance x Quality, shift KPIs, and a downtime pareto (which reasons cost the most time).

Where the data comes from (real factory). Computed from production records, downtime logs and shifts — the headline OEE metric of any MES.

How to connect. Fully derived — nothing to enter.

Timeline

What it does. A chronological feed of machine status-change events (Running to Breakdown, etc.).

Where the data comes from (real factory). Every time a machine changes state, the change is logged.

How to connect. Automatic — whenever a machine changes state, whether by a person or a PLC signal.

3. Operations Pack

Planning and running production jobs end to end.

Work Orders

What it does. Production jobs: part number, batch, target quantity, assigned machine, deadline and status (Planned to In Progress to Completed). Marking a work order Completed automatically issues the raw material it consumed and adds the finished goods to stock, using the Bill of Materials (BOM).

Where the data comes from (real factory). In a real factory a work order originates in the ERP or planning system, or from a confirmed customer/sales order; the planner then releases it to the floor.

How to connect. Three ways: (1) create it manually here; (2) sync from your ERP (Tally / SAP / Excel) via CSV import or an API integration; (3) auto-generate it from a customer order. The BOM-driven stock deduction then happens on its own.

Production Plan

What it does. Assigns work orders to specific machines, shifts and dates, tracking planned vs actual quantity so you can spot slippage early.

Where the data comes from (real factory). The production planning team.

How to connect. Manual, built on top of work orders.

Scheduling

What it does. A calendar view of all jobs across machines and shifts; reassign and reschedule when a machine goes down for maintenance.

Where the data comes from (real factory). The planning team.

How to connect. Manual scheduling; reflects work orders and machine availability.

Operator Terminal

What it does. Where a shop-floor operator starts and finishes a job at the machine and logs good and rejected piece counts.

Where the data comes from (real factory). Operators on a tablet or terminal at the line. In more automated lines, the good/reject counts come from the machine's own part counter via the PLC.

How to connect. Manual entry by operators, or automatic counts pulled from the PLC through the edge agent.

Orders and Dispatch

What it does. Customer orders and dispatch/shipment tracking, linked back to the work orders that fulfil them.

Where the data comes from (real factory). Sales, ERP or CRM.

How to connect. Manual entry, or integrate with your sales/ERP system.

4. Factory Pack

Full shop-floor visibility — materials, quality and maintenance.

Maintenance AI

What it does. Predictive-maintenance risk scoring per machine, flagging machines likely to need attention.

Where the data comes from (real factory). Computed from each machine's downtime history and vibration/telemetry.

How to connect. Derived from the machine and IoT data already in FlowMES.

CMMS

What it does. Maintenance management: preventive and breakdown tasks, schedules, spare parts used and downtime per task, building a service history per machine.

Where the data comes from (real factory). The maintenance team.

How to connect. Manual; preventive schedules automatically raise upcoming tasks.

Quality

What it does. Inspections logged against each work order: quantity inspected, passed, failed, defect category, rework and scrap — full batch traceability.

Where the data comes from (real factory). QC inspectors at inspection stations.

How to connect. Manual entry; in advanced setups, results can come from digital gauges or machine-vision systems.

Inventory

What it does. Live stock of raw materials, spares and finished goods, with every receive / issue / return / adjustment recorded and low-stock alerts. For a compressor manufacturer like GMATS it adds the enterprise model: 4-bucket stock (Physical / Reserved / Available), item aliases (many names to one item), Proforma reservation, Tax-Invoice deduction (with a printable GST invoice), free-spares issue notes, cycle counts and a variance report.

Where the data comes from (real factory). The store and purchase teams. Opening stock is imported from Tally.

How to connect. Manual entry; Tally/Excel CSV import; automatic deduction when a work order completes (BOM); and the full Proforma to Tax-Invoice flow for sales.

Purchasing

What it does. Purchase orders and a supplier master; receiving a PO updates stock and creates a receipt transaction.

Where the data comes from (real factory). The purchase team.

How to connect. Manual; can integrate with supplier portals or ERP.

Digital Twin

What it does. A live map of the factory floor showing machine positions and their real-time status.

Where the data comes from (real factory). Machine coordinates configured once, plus live status.

How to connect. Lay the machines out once; status then streams in from the IoT / Connectivity layer.

5. Intelligence Pack

Live machine data, AI and connectivity.

IoT Command

What it does. Live machine telemetry — temperature, vibration, spindle speed, power — refreshed every few seconds, with alerts when a machine runs outside safe limits.

Where the data comes from (real factory). Sensors on the machines.

How to connect. An edge device on your network reads the sensors and publishes the readings over MQTT to FlowMES.

AI Insights

What it does. Rule-based recommendations (schedule a bearing inspection, change tooling, lubrication overdue) each with a confidence score.

Where the data comes from (real factory). Computed from machine, quality and downtime patterns.

How to connect. Derived automatically.

AI Copilot

What it does. Ask questions about your factory in plain English ('why is OEE low?', 'what should I reorder?'), get AI root-cause analysis and a one-click daily management report — powered by Claude.

Where the data comes from (real factory). It reads the live data already in FlowMES; nothing extra to enter.

How to connect. Off by default. To switch on: add ANTHROPIC_API_KEY in Railway (optionally AI_MODEL, default a cheap fast model). That single environment variable is the only step.

Connectivity

What it does. The industrial protocol adapter layer — OPC UA, Modbus TCP, Siemens S7, Allen-Bradley, Beckhoff and Omron — listing connected PLCs and their live signals.

Where the data comes from (real factory). The PLCs that control your machines.

How to connect. Run the FlowMES edge agent on a small PC on your shop-floor network. It speaks each PLC's native protocol (using the vendor library), reads the tags/registers, normalises them and pushes them to FlowMES over an outbound connection. See Section 7.

Executive OEE

What it does. Management-level OEE dashboards and trends for the plant.

Where the data comes from (real factory). Production records.

How to connect. Derived.

Escalations

What it does. Automatic alerts — low stock, breakdown, overdue work order, OEE below target — routed to an owner and department so nothing slips.

Where the data comes from (real factory). Rules running over the live data.

How to connect. Auto-generated; can also be raised manually.

Notifications

What it does. A system notification feed (alerts, warnings, info) for the team.

Where the data comes from (real factory). Events across the system.

How to connect. Automatic.

6. Admin Pack

Management, compliance and the SaaS platform layer.

Documents

What it does. Controlled documents — SOPs, quality plans, maintenance plans, compliance certificates — with version, owner and review-due date.

Where the data comes from (real factory). Your document-control process.

How to connect. Manual entries / links (e.g. links to cloud storage).

SaaS Admin

What it does. Tenant / company management for the platform: plan, seats, subscription status and monthly fee.

Where the data comes from (real factory). You, the platform operator.

How to connect. Platform-owner view.

User Management

What it does. Add employees and assign roles (Admin / Supervisor / Operator) and reset passwords — scoped to your own company. Self-registration is disabled; only an Admin can add people.

Where the data comes from (real factory). Your company's Admin.

How to connect. The Admin adds users in-app; each new user can only sign in with the role assigned.

Costing

What it does. Cost records (material, labour, overhead) linked to jobs, for basic cost visibility.

Where the data comes from (real factory). Finance / accounts.

How to connect. Manual; can be derived from production and inventory movements.

Enterprise Polish

What it does. The platform plumbing: audit log (who did what, when), system health, an executive summary and report requests.

Where the data comes from (real factory). The system itself.

How to connect. Automatic.

7. How real machine data gets into FlowMES

This is the question every factory asks. The path is the same whether the machine speaks MQTT or a PLC protocol:



The edge agent — what it is

A lightweight program on a small industrial PC or Raspberry Pi sitting on your factory's local network. It reads from your machines two ways: (1) modern machines that already publish data — it subscribes over **MQTT**; (2) PLC-controlled machines — it speaks the PLC's protocol directly (OPC UA, Modbus, Siemens S7, Allen-Bradley, Beckhoff, Omron) using that protocol's library. It then sends the readings to the FlowMES cloud.

"But our PLCs don't allow external connections"

That's exactly why the edge agent exists, and it's the common case. You do **not** open any port on the PLC to the internet. The edge agent talks to the PLC **locally**, and makes only **outbound** connections to FlowMES — the same direction your web browser uses. The PLC never sees external traffic. If a machine has no digital output at all, simple add-on sensors (temperature, current, vibration) feed the edge agent instead — no change to the machine.

Latency and safety

For monitoring and dashboards, data is live within 1-3 seconds. Safety-critical control (e.g. auto-shutdown on overpressure) stays in the local PLC/edge layer so it responds in milliseconds without depending on the cloud. The cloud handles analytics, inventory, work orders and reporting — not real-time safety control.

8. Security, multi-company and running it

Security and access

Passwords are stored with bcrypt. All traffic is over HTTPS. Access is role-based — Admin (full), Supervisor (create/approve), Operator (update only) — enforced on the server, not just hidden in the UI. Each company's data is **isolated by tenant** and enforced on the backend, so one client can never see another's data.

Multi-company (white-label SaaS)

FlowMES is multi-tenant: each customer is a 'company' with its own data, its own logins, its own branding (name, logo, colour) and its own licensed module packs (Starter / Growth / Enterprise). Onboarding a new client is a checklist — create the company, pick its plan, import or seed its data, create its admin login — no code change.

Where it's deployed

Frontend on **Vercel** (flow-mes.vercel.app), backend + PostgreSQL on **Railway** (flowmes-production.up.railway.app). Both auto-deploy from the Git repository. HTTPS is automatic on both.

How to run it

Use it now: open flow-mes.vercel.app and sign in (it's always live). **Run locally for development:** start the backend (FastAPI on port 8000, needs a PostgreSQL DATABASE_URL) and the frontend (Next.js on port 3000) in two terminals; the frontend is already pointed at the local backend.

FlowMES — one system for machines, materials, production and quality, built for the way Indian SME manufacturers actually work.